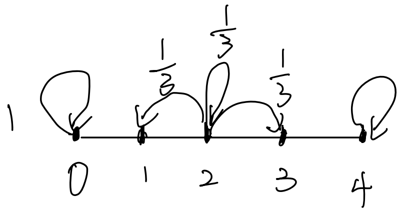
4.1



一步转移概率矩阵：

$$P=\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

两步：$$P^{(2)}=PP=\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{4}{9} & \frac{2}{9} & \frac{2}{9} & \frac{1}{9} & 0 \\ \frac{1}{9} & \frac{2}{9} & \frac{1}{3} & \frac{2}{9} & \frac{1}{9} \\ 0 & \frac{1}{9} & \frac{2}{9} & \frac{2}{9} & \frac{4}{9} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

4.2

$$P^{(1)}=\begin{pmatrix} p & 1-p & 0 & 0 \\ 0 & 0 & p & 1-p \\ p & 1-p & 0 & 0 \\ 0 & 0 & p & 1-p \end{pmatrix} \qquad P^{(2)}=\begin{pmatrix} p^2 & p(1-p) & p(1-p) & (1-p)^2 \\ p^2 & p(1-p) & p(1-p) & (1-p)^2 \\ p^2 & p(1-p) & p(1-p) & (1-p)^2 \\ p^2 & p(1-p) & p(1-p) & (1-p)^2 \end{pmatrix}$$

4.3

(1) $P\{X_{n+1}=i_{n+1}, X_{n+2}=i_{n+2}, \cdots, X_{n+m}=i_{n+m} \mid X_0=i_0, X_1=i_1, \cdots, X_n=i_n\}$

$= P\{X_{n+2}=i_{n+2}, \cdots, X_{n+m}=i_{n+m} \mid X_{n+1}=i_{n+1}, X_0=i_0, X_1=i_1, \cdots, X_n=i_n\}$
$\quad P\{X_{n+1}=i_{n+1} \mid X_0=i_0, X_1=i_1, \cdots, X_n=i_n\}$

$= P\{X_{n+2}=i_{n+2}, \cdots, X_{n+m}=i_{n+m} \mid X_{n+1}=i_{n+1}, X_0=i_0, X_1=i_1, \cdots, X_n=i_n\}$
$\quad P\{X_{n+1}=i_{n+1} \mid X_n=i_n\}$

$= P\{X_{n+3}=i_{n+3}, \cdots, X_{n+m}=i_{n+m} \mid X_{n+2}=i_{n+2}, X_0=i_0, \cdots, X_{n+1}=i_{n+1}\}$
$\quad P\{X_{n+2}=i_{n+2} \mid X_{n+1}=i_{n+1}\} P\{X_{n+1}=i_{n+1} \mid X_n=i_n\}$

$= P\{\underline{X_{n+m}=i_{n+m}} \mid X_0=i, X_1=i_1 \cdots, \underline{X_{n+m-1}=i_{n+m-1}}\}$
$\quad P\{X_{n+1}=i_{n+1}, \cdots, X_{n+m-1}=i_{n+m-1} \mid X_n=i_n\}$

得证

$= P\{X_{n+m}=i_{n+m} \mid X_{n+m-1}=i_{n+m-1}\} P\{X_{n+1}=i_{n+1}, \cdots, X_{n+m-1}=i_{n+m-1} \mid X_n=i_n\}$

(2) 原式 $= P\{X_{n+2}=i_{n+2},\cdots,X_{n+m}=i_{n+m} \mid X_0=i_0,\cdots,X_{n+1}=i_{n+1}\}$

$P\{X_0=i_0,\cdots,X_n=i_n \mid X_{n+1}=i_{n+1}\}$ 由(1)结论

$= P\{X_{n+2}=i_{n+2},\cdots,X_{n+m}=i_{n+m} \mid X_{n+1}=i_{n+1}\}$

$P\{X_0=i_0,\cdots,X_n=i_n \mid X_{n+1}=i_{n+1}\}$ 得证

4.4.

$P\{X_2=4 \mid 1<X_1<4\} = P\{X_2=4 \mid X_1=2\} + P\{X_2=4 \mid X_1=3\}$

已知初始分布 $p_0 = (\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})^T$

$$P^{(2)} = \begin{pmatrix} \frac{1}{4} & \frac{7}{32} & \frac{1}{4} & \frac{9}{32} \\ \frac{1}{4} & \frac{7}{32} & \frac{1}{4} & \frac{9}{32} \\ \frac{1}{4} & \frac{7}{32} & \frac{1}{4} & \frac{9}{32} \\ \frac{1}{4} & \frac{7}{32} & \frac{1}{4} & \frac{9}{32} \end{pmatrix} \qquad p_2 = p_0^T P^{(2)}$$

则 $P\{X_2=4 \mid 1<X_1<4\} =$

$$P\{X_2=4 \mid X_0=1, 1<X_1<4\} = \frac{P\{X_0=1, 1<X_1<4, X_2=4\}}{P\{X_0=1, 1<X_1<4\}} = \frac{P_1P_{12}P_{24}+P_1P_{13}P_{34}}{P_1P_{12}+P_1P_{13}}$$

$$= \frac{\frac{1}{4}\times\frac{1}{4}\times\frac{1}{4}+\frac{1}{4}\times\frac{1}{4}\times\frac{3}{8}}{\frac{1}{4}\times\frac{1}{4}+\frac{1}{4}\times\frac{1}{4}}$$

$$= \frac{5}{16}$$

$P\{X_2=4 \mid 1<X_1<4\} = \frac{19}{60}$ 同理

4.5 证 $P\{Y_n \mid Y_{n-1},Y_{n-2},\cdots Y_0\} = P\{Y_n \mid Y_{n-1}\}$

$Y_n = X_n - cY_{n-1}$，Y_n是$(X_1,X_2,\cdots,X_n)$的函数，又$X_1,X_2,\cdots X_n\cdots$独立同分布

对$\forall n\geq 0$，X_{n+1}与$(Y_0,Y_1,\cdots,Y_n)$独立.

$P\{Y_{n+1}=i_{n+1} \mid Y_0=0,\cdots,Y_n=i_n\}$

$= P\{Y_{n+1}+cY_n=i_{n+1}+ci_n \mid Y_0=0,Y_1=i_1,\cdots,Y_n=i_n\}$

$= P\{X_{n+1}=i_{n+1}+ci_n \mid Y_0=0,\cdots,Y_n=i_n\}$

$= P\{X_{n+1}=i_{n+1}+ci_n\} = P\{X_{n+1}=i_{n+1}+ci_n \mid Y_n=i_n\} = P\{Y_{n+1}=i_{n+1} \mid Y_n=i_n\}$.

4.6. $P^{(3)} = PPP = \begin{pmatrix} 0.25 & 0.375 & 0.375 \\ 0.375 & 0.25 & 0.375 \\ 0.375 & 0.375 & 0.25 \end{pmatrix}$

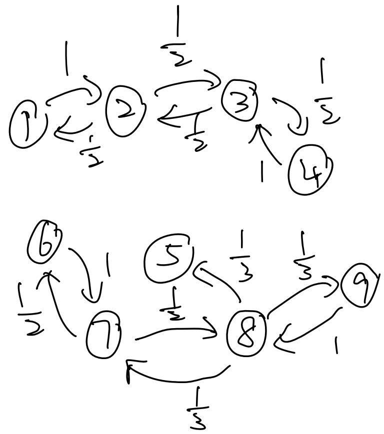
$$p(3) = p(0)^T P^{(3)} = (0,\ 0,\ 1)\ P^{(3)} = (0, 0, 0.25)$$

$\therefore p_3(3) = 0.25$

4.7. 略

4.8. 略

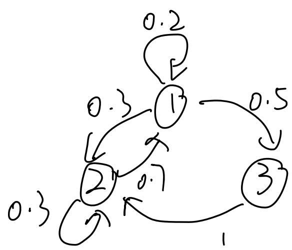
4.9.

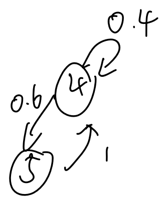


2闭集

4.10

(1)

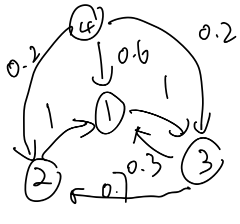




$C_1 = \{1,2,3\}$ 和 $C_2 = \{4, 5\}$

正常返遍历的两个闭集

(2)



$C_1 = \{1, 2, 3\}$ 遍历闭集，$N = \{4\}$ 非常返

(3) $C_1 = \{1, \cdots, b-1\}$ 非常返闭集，$C_2 = \{0, b\}$ 吸收态闭集

4.11. (1)

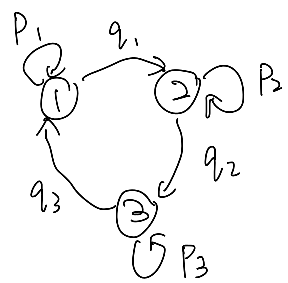
$$f_{11}^{(n)} = \begin{cases} \frac{1}{2}, & n=1 \\ \frac{1}{6}, & n=2 \\ \frac{1}{6}\left(\frac{2}{3}\right)^{n-2}, & n>2 \end{cases}$$

$$f_{12}^{(n)} = \begin{cases} \frac{1}{2}, & n=1 \\ \frac{1}{4}, & n=2 \\ \left(\frac{1}{2}\right)^{n-2}\frac{1}{4}, & n>2 \end{cases}$$

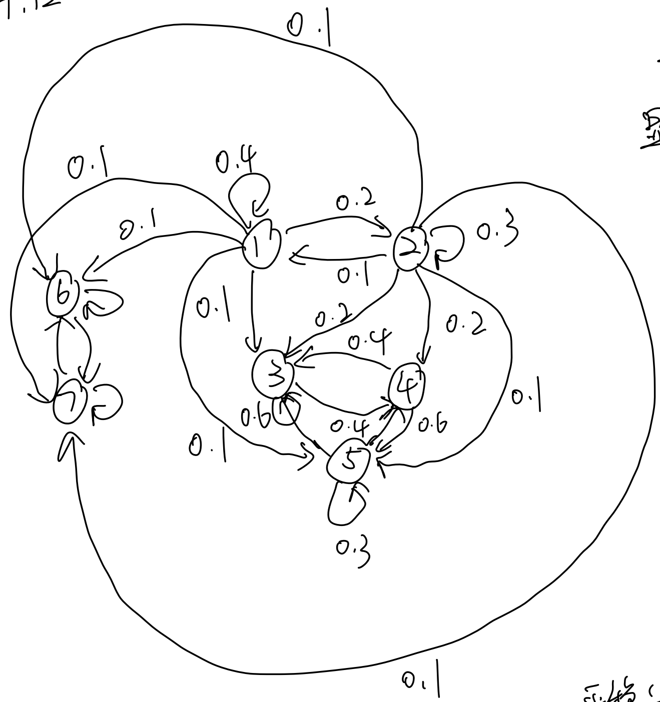
(2)

$$f_{11}^{(n)} = \begin{cases} p_1, & n=1 \\ \sum_{k=0}^{n-3} q_1 q_2 q_3 (p_2)^k (p_3)^{n-3-k}, & n\geq 3 \end{cases}$$

$$f_{12}^{(n)} = \begin{cases} q_1, & n=1 \\ (p_1)^{n-1} q_1, & n\geq 2 \end{cases}$$



4.12



$f_{11}^{(1)} = 0.4$, $f_{11}^{(2)} = 0.02$, $f_{11}^{(n)} = 0.02(0.3)^{n-2}$

显然 $f_{11} = \sum_{k=1}^{\infty} f_{11}^{k} < 1$

有非常返集 $D = \{1, 2\}$

正常返非遍历闭集 $C_1 = \{3, 4, 5\}$ 和 $C_2 = \{6, 7\}$

C_1 转移矩阵：

$$P_1 = \begin{pmatrix} 0.6 & 0.4 & 0 \\ 0.4 & 0 & 0.6 \\ 0.2 & 0.5 & 0.3 \end{pmatrix}$$

平稳分布 $\pi = \pi P_1$

$$\begin{cases} 0.6\pi_1 + 0.4\pi_2 + 0.2\pi_3 = \pi_1 \\ 0.4\pi_1 + 0.5\pi_3 = \pi_2 \\ 0.6\pi_2 + 0.3\pi_3 = \pi_3 \end{cases}$$

∴平稳分布 $\{0, 0, \frac{10}{23}, \frac{7}{23}, \frac{6}{23}, 0, 0\}$

同理 C_2 平稳分布 $\{0, 0, 0, 0, 0, \frac{8}{15}, \frac{7}{15}\}$

4.13

$$\begin{cases} \pi_1 q_1 = \pi_0 \\ \pi_0 + q_2 \pi_2 = \pi_1 \\ \pi_1 p_1 + q_3 \pi_3 = \pi_2 \\ \cdots \\ \pi_{n-1} p_{n-1} + q_{n+1} \pi_{n+1} = \pi_n \end{cases}$$

解得 $\pi_j = \dfrac{p_1 \cdots p_{j-1}}{q_1 \cdots q_j} \pi_0, \quad j \geq 1$

$$\pi_0 = \frac{1}{1 + \sum_{j=1}^{\infty} \prod_{k=0}^{j-1} \frac{p_k}{q_{k+1}}}$$

4.14

$P_{ii} = 0 \qquad P_{i(i+1)} = \dfrac{2N-i}{2N} \qquad P_{j(i-1)} = \dfrac{j}{2N} \quad , j = 0, 1, 2, \cdots, 2N$

$\{\pi_j, j = 0, 1, \cdots, 2N\}$ 满足

$$\begin{cases} \pi_0 = \frac{1}{2N} \pi_1 \\ \pi_j = \pi_{j-1} \frac{2N-j+1}{2N} + \pi_{j+1} \frac{j+1}{2N}, \quad 1 \leq j \leq 2N-1 \\ \pi_{2N} = \frac{1}{2N} \pi_{2N-1} \end{cases}$$

解得 $\pi_j = C_{2N}^j \pi_0$，又 $\sum_j \pi_j = 1$，$\pi_0 = 2^{-2N}$

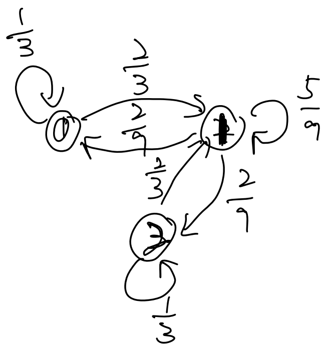
$\therefore$ 平稳分布 $\pi_j = C_{2N}^j 2^{-2N}$，$j = 0, 1, 2, \cdots, 2N$.

4.15

(1)

$$P=\begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 \\ \frac{2}{9} & \frac{5}{9} & \frac{2}{9} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{pmatrix}$$

(2)



首先状态集 $I=\{0,1,2\}$ 有限
且不可约马尔可夫链（互通），
∴该链状态全为正常返，
又1的周期为 $d=1$，∴得证。

(3) 由于(2)中已证得该链为非周期不可约马尔可夫链，且为遍历链（正常返）

$\Rightarrow$ 存在平稳分布 $\{\pi_i,\ 0\le i\le 2\}$

$$\begin{cases} \frac{1}{3}\pi_0+\frac{2}{9}\pi_1=\pi_0 \\ \frac{2}{3}\pi_0+\frac{5}{9}\pi_1+\frac{2}{3}\pi_2=\pi_1 \\ \frac{2}{9}\pi_1+\frac{1}{3}\pi_2=\pi_2 \end{cases}$$

解得平稳分布 $\{\frac{1}{5},\frac{3}{5},\frac{1}{5}\}$

又 $\lim\limits_{n\to\infty}P_{ij}^{(n)}=\frac{1}{\mu_j}=\pi_j$，$j=0,1,2$，易得

$\lim\limits_{n\to\infty}P_{i0}^{(n)}=\frac{1}{5}$ $\quad\lim\limits_{n\to\infty}P_{i1}^{(n)}=\frac{3}{5}$ $\quad\lim\limits_{n\to\infty}P_{i2}^{(n)}=\frac{1}{5}$

4.16 (1) 归纳法，当 $n=m$ 时，对 $\forall j\in I$，$\sum\limits_{i\in I}P_{ij}^{(m)}=1$，则

$$\sum_{j\in I}P_{ij}^{(m+1)}=\sum_{i\in I}\underbrace{\left(\sum_{k\in I}P_{ik}^{(m)}P_{kj}^{(1)}\right)}_{\text{CK方程}}=\sum_{k\in I}P_{kj}^{(1)}\left(\sum_{i\in I}P_{ik}^{(m)}\right)=\sum_{k\in I}P_{kj}^{(1)}=1$$

∴得证

(2) 由条件知 $\{X(n),\ n\ge 1\}$ 为非周期不可约状态空间有限马尔可夫链，

∴$\{X(n),\ n\ge1\}$ 为遍历链，

即 $\lim\limits_{n\to\infty}P_{ij}^{(n)}=\frac{1}{\mu_j}=\pi_j>0$，$j\in I$

各状态平均返回时间为 μ_j，∵ $\sum\limits_{m\in I}\pi_m=\sum\limits_{m\in I}\lim\limits_{n\to\infty}P_{im}^{(n)}=\sum\limits_{m\in I}\frac{1}{\mu_m}=1$

$\therefore m \cdot \frac{1}{\mu m} = 1$ 即 $\mu_j = m$, $j = 1, 2, \dots, m$